

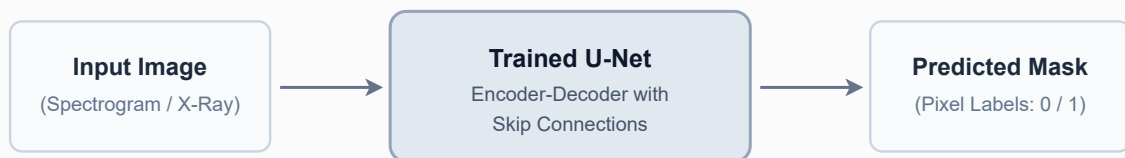
U-Net: Convolutional Networks for Image Segmentation

Introduction & Applications

U-Net is a specialized convolutional neural network architecture originally developed in 2015 by Olaf Ronneberger et al. for biomedical image segmentation. Unlike standard classification networks that assign a single label to an entire image, U-Net performs **pixel-wise semantic segmentation**, outputting a precise mask that pinpoints both the boundary, shape, and spatial location of target objects.

- **Key Applications:**
 - **Satellite & Remote Sensing:** Automatically delineating roads, buildings, water bodies, and vegetation zones from aerial imagery.
 - **Medical Imaging:** Detecting and localizing tumors, lesions, or organ boundaries in X-rays, Ultrasounds, MRIs, and CT scans.
- **Mask & Ground Truth Representation:**
 - **Binary Segmentation:** Images contain two classes: Background (0) and Foreground / Target (1).
 - **Multi-Class Segmentation:** Each target entity is assigned a unique integer (0 = background, 1 = tumor, 2 = vessel, 3 = organ).

FIGURE 1: IMAGE CLASSIFICATION VS. PIXEL-WISE SEMANTIC SEGMENTATION



Mathematical Step-by-Step Operations (Toy 5×5 Example)

To understand the computational mechanics of U-Net, consider a single-channel normalized image matrix (5×5 pixels with values scaled to [0, 1]):

1. Convolution & Feature Maps

- **Filter / Kernel:** A matrix of trainable weights plus a bias weight b optimized during gradient descent.
- **Dot Product Operation:** Overlays the filter on the image patch, computes the element-wise product sum, and adds bias b :
$$\text{Feature Map Value} = (\sum W_{ij} \cdot X_{ij}) + b$$
- **Stride:** The step size the kernel moves across the image (stride = 1).
- **Padding:** Adding zero-value borders around the input matrix to ensure the resulting feature map maintains the exact spatial resolution.

2. Activation Function (ReLU)

- **Rectified Linear Unit:** Computes $f(z) = \max(0, z)$. Clamps all negative outputs to zero while passing positive activations through unaltered, introducing crucial non-linear representation.

3. Downsampling (Max Pooling)

- **Operation:** Applies a 2×2 sliding window with a stride of 2 across the feature map, extracting only the maximum numerical value in each window.
- **Purpose:** Halves spatial height and width (H/2, W/2), reducing computational overhead while expanding the receptive field to capture global contextual patterns.

4. Upsampling Techniques

- **Nearest Neighbor Upsampling:** Copies each scalar value from the pooled matrix into a 2×2 block. It is non-parametric (contains no trainable weights).
- **Transposed Convolution (Deconvolution):** Utilizes trainable filter weights and a bias term to mathematically project low-resolution feature maps back to higher spatial resolutions, learning optimal reconstruction patterns.

FIGURE 2: DOWNSAMPLING (MAX POOLING) AND UPSAMPLING (TRANSPPOSED CONV)



5. Skip Connections & Final Probability Head

- **Skip Connections:** Directly concatenate fine-grained encoder feature maps with upsampled decoder feature maps. This compensates for spatial coordinate loss caused by pooling.
- **1×1 Convolution Head:** Condenses multi-channel features into the target class count.
- **Sigmoid Activation & Thresholding:**

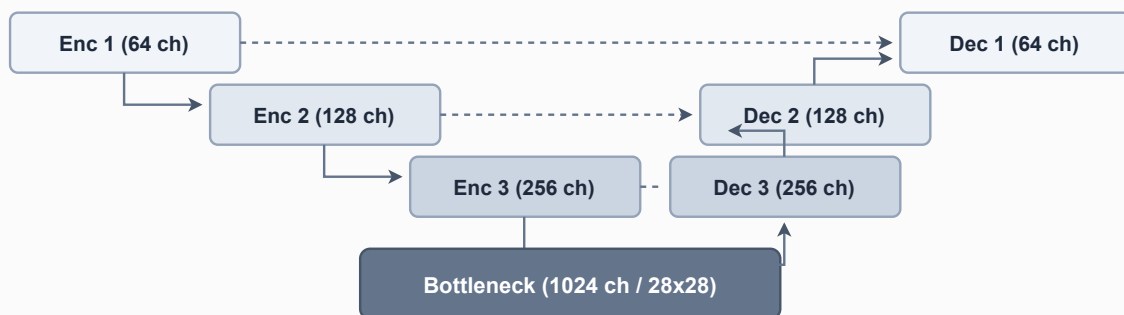
$$\sigma(z) = 1 / (1 + e^{-z})$$

$$\text{Class}(x, y) = 1 \text{ if } \sigma(z) \geq 0.5 \text{ else } 0$$

The Complete U-Net Architecture (Ronneberger et al. 2015)

The network forms a symmetric U-shape with an **Encoder (Contracting Path)** to extract context and a **Decoder (Expansive Path)** to enable precise localization.

FIGURE 3: U-NET SYMMETRIC ARCHITECTURE & SKIP CONNECTIONS



Key Architectural Properties:

- **Weight Parameter Calculations:**
 - First Conv Layer (3×3×1 input, 64 filters): $64 \times (3 \times 3 \times 1 + 1) = \mathbf{640 \text{ parameters}}$.
 - Second Conv Layer (3×3×64 inputs, 64 filters): $64 \times (3 \times 3 \times 64 + 1) = \mathbf{36,928 \text{ parameters}}$.
 - Total Network Parameters: Approximately **31.03 Million**.
- **Valid vs Same Padding:** The original paper used unpadded convolutions (requiring cropping during skip- concatenation). Modern implementations utilize padding='same' to preserve identical spatial dimensions throughout.

Segmentation Evaluation Metrics

Metric	Mathematical Formula	Strengths / Limitations
Pixel Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Misleading for sparse targets. If background is 95% of the image, predicting all background yields 95% accuracy while completely missing the target.
Intersection over Union (IoU / Jaccard)	$ A \cap B / A \cup B = TP / (TP + FP + FN)$	Standard benchmark metric. Directly measures spatial bounding overlap (1.0 = perfect match).
Dice Coefficient (F1 Score)	$2 A \cap B / (A + B) = 2TP / (2TP + FP + FN)$	Heavily penalizes false positives and negatives; tends to yield slightly higher numerical scores than IoU.

Python / Keras Implementation & Training

1. Data Loading & Multi-Mask Fusion

```
# Resize to 128x128, grayscale conversion, and min-max normalization (/ 255.0)
# Multiple masks for the same patient are combined using element-wise addition:
if is_mask:
    mask = (img > 0.5).astype(np.float32)
    current_mask = np.clip(current_mask + mask, 0.0, 1.0)
```

2. Weight Initialization (He Normal / Kaiming)

- **Formula:** Drawn from Normal distribution with mean = 0 and standard deviation:
 $\sigma = \sqrt{2 / n_{in}}$, where $n_{in} = k_w \times k_h \times c_{in}$
- **Impact:** Utilizing kernel_initializer='he_normal' boosted the test IoU score from ~0.60 to ~0.70 across 40 epochs.

3. Modern Enhancements

- **Residual U-Net (Res-UNet):** Adds residual skip additions inside convolution blocks to prevent vanishing gradients in deep networks.
- **Attention U-Net:** Incorporates attention gates on skip connections to filter out irrelevant background noise and focus activations on salient target regions.